

STM 32 BASED PID TEMPERATURE CONTROLLER

INTERNSHIP REPORT

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In partial fulfilment of the requirement for award of

B.E IN ELECTRONICS AND INSTRUMENTATION ENGINEERING



DEPARTMENT OF ELECTRONICS AND INSTRUMENTAION ENGINEERING

KUMARAGURU COLLEGE OF TECHNOLOGY

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COIMBATORE – 641 049

CERTIFICATE



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This is to certify that **following students** has successfully completed an internship at **Transzio® Integral Systems LLP** from **13/12/2025** to **13/01/2026**.

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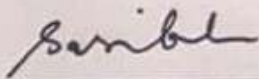
During the internship period, **they** worked as a **Engineer Trainee** and demonstrated sincerity, dedication, and a strong willingness to learn and their performance and conduct were found to be satisfactory.

We found **them** to be hardworking and responsible. We wish **them** every success in future endeavours.

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ABSTRACT



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This internship focuses on the design and development of a milk temperature monitoring and control system using the STM32 microcontroller. The work involved two major implementations: a temperature monitoring with ON/OFF control system and a PID-based temperature control system. An NTC thermistor was used to measure temperature, and the sensed data was processed using the STM32 ADC. The Steinhart–Hart equation was implemented to accurately convert thermistor resistance into temperature values. For temperature regulation, both ON/OFF control logic and PID control algorithms were developed and tested. The system demonstrated stable temperature control, improved accuracy, and efficient response to temperature variations. This internship provided practical exposure to embedded systems, sensor interfacing, control algorithms, and real-time system implementation using STM32.

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CHAPTER 1



CHAPTER 1

INTRODUCTION

1.1 Introduction

Temperature monitoring and control are critical requirements in food processing industries, especially in applications involving **milk heating and pasteurization**. Maintaining milk at a precise temperature is essential to ensure product quality, safety, and nutritional value. Overheating can degrade milk proteins, while underheating may lead to microbial growth. Hence, accurate and reliable temperature control systems are necessary.

With the advancement of embedded systems, microcontroller-based temperature control solutions have become widely adopted due to their accuracy, flexibility, and real-time monitoring capability. This internship project focuses on the **design and implementation of a milk temperature monitoring and control system using an STM32 microcontroller**. The system ensures controlled heating of milk by continuously measuring temperature and regulating the heater using appropriate control algorithms.

1.2 Background of the Project

Milk heating processes are commonly used in dairy applications such as:

- Milk pasteurization
- Milk boiling and warming
- Dairy product preparation
- Small-scale and automated milk processing units

Traditional milk heating systems often rely on manual supervision or simple thermostats, which can lead to temperature fluctuations and inconsistent heating. To overcome these limitations, an automated embedded control system is required.

In this project, an **NTC thermistor** is used to sense the milk temperature, and the **STM32 microcontroller** processes the sensor data using its high-resolution ADC. Based on the measured temperature, the system controls the heating element to maintain the desired setpoint.

1.3 Existing System

The existing milk heating systems generally use:

- Manual ON/OFF switches
- Analog thermostats

- Relay-based temperature controllers

These systems have several disadvantages:

- Poor temperature accuracy
- Large overshoot during heating
- No digital monitoring or data logging
- Limited adaptability to temperature variations
- High dependency on human intervention

Such limitations can negatively affect milk quality and process efficiency.

1.4 Proposed System

The proposed system is an **STM32-based Milk Temperature Monitoring and Control System** designed to overcome the limitations of conventional methods. The system continuously monitors milk temperature using an NTC thermistor and controls the heating element using digital control logic.

The project is implemented in two stages:

1.4.1 Temperature Monitoring and ON/OFF Control

- Real-time milk temperature measurement
- Digital temperature display via UART
- Heater control using ON/OFF logic with hysteresis
- Basic protection against overheating

1.4.2 PID-Based Milk Temperature Control

- Implementation of Proportional–Integral–Derivative (PID) control
- Reduced temperature fluctuation
- Improved steady-state accuracy
- Smooth and controlled heating of milk

This approach ensures stable temperature control suitable for dairy applications.

1.5 Objective of the Project

The objectives of this internship project are:

- To design a milk temperature monitoring system using STM32
- To interface an NTC thermistor with STM32 ADC

- To implement ON/OFF temperature control for milk heating
- To analyze the limitations of ON/OFF control
- To design and implement a PID-based milk temperature controller
- To compare ON/OFF and PID control performance
- To gain practical experience in embedded control systems

1.6 Scope of the Project

The scope of the project includes:

- Embedded system design for milk heating applications
- Accurate temperature measurement using NTC thermistor
- Digital control of heating element
- Serial communication for monitoring
- System analysis and tuning of control parameters

Future enhancements may include:

- LCD or OLED temperature display
- Automatic milk pasteurization cycles
- Data logging and quality tracking
- IoT-based remote monitoring
- Advanced adaptive PID control

1.7 Conclusion

This chapter is about the introduction to milk temperature monitoring and control systems. It explains the importance of maintaining proper temperature during milk heating to ensure quality and safety. The need for automation using embedded systems is discussed. The role of the STM32 microcontroller in temperature measurement and control is highlighted. This chapter provides the basic motivation and objective of the project.

CHAPTER 2



CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

The literature survey provides an overview of existing research and technological developments related to **temperature monitoring, milk heating systems, and control techniques**. It highlights various methods used for temperature sensing, heating control, and automation in food processing industries. Special emphasis is given to **ON/OFF control systems** and **PID-based temperature control**, which form the foundation of this internship project.

2.2 Temperature Control in Dairy Applications

Milk heating is a critical process in dairy industries for applications such as pasteurization, sterilization, and product preparation. Maintaining milk within a specific temperature range is essential to preserve nutritional quality and prevent microbial contamination.

Several studies have emphasized that:

- Temperature deviations during milk heating can degrade proteins and fats.
- Uniform and controlled heating improves product consistency.
- Automated temperature control systems reduce human error.

Traditional dairy heating systems often lack precise control, leading to uneven heating and energy inefficiency.

2.3 Temperature Sensing Techniques

Various temperature sensors are used in industrial and embedded applications, including:

2.3.1 Thermocouples

Thermocouples are widely used for high-temperature measurements. However, they have lower sensitivity at moderate temperature ranges and require cold-junction compensation.

2.3.2 Resistance Temperature Detectors (RTDs)

RTDs provide high accuracy but are expensive and require precise signal conditioning circuits.

2.3.3 NTC Thermistors

NTC (Negative Temperature Coefficient) thermistors are commonly used in milk heating systems due to:

- High sensitivity in moderate temperature ranges
- Low cost
- Fast response time
- Ease of interfacing with microcontrollers

Because of these advantages, NTC thermistors are widely preferred for embedded temperature control applications and are used in this project.



Fig.2.1 NTC Thermistor

2.4 Microcontroller-Based Temperature Monitoring Systems

Recent advancements in microcontroller technology have enabled the development of compact and reliable temperature control systems. Microcontrollers offer:

- High-speed data processing
- Integrated ADC modules
- Real-time control capability
- Programmability and flexibility

Several studies have demonstrated that microcontroller-based systems provide better accuracy and reliability compared to analog controllers. Digital systems also allow easy modification of control algorithms without hardware changes.

2.5 STM32 Microcontroller in Control Applications

The STM32 microcontroller series is widely used in industrial and academic projects due to:

- 32-bit ARM Cortex architecture
- High-resolution ADC
- Low power consumption
- Rich peripheral support

Literature indicates that STM32 microcontrollers are suitable for precise temperature measurement and real-time control applications. Their ADC accuracy makes them ideal for interfacing with analog sensors such as NTC thermistors.

2.6 ON/OFF Temperature Control Technique

ON/OFF control is the simplest form of temperature control. In this method:

- The heater is turned ON when the temperature falls below the setpoint
- The heater is turned OFF when the temperature exceeds the setpoint

Although ON/OFF control is easy to implement, several studies highlight its drawbacks:

- Temperature oscillations around the setpoint
- Frequent switching of the heater
- Reduced control accuracy

To reduce oscillations, hysteresis is commonly introduced. ON/OFF control is suitable for basic milk heating applications but may not meet high-precision requirements.

2.7 PID Temperature Control Technique

Proportional–Integral–Derivative (PID) control is one of the most widely used control techniques in industrial applications. PID controllers offer:

- Fast response (Proportional action)
- Zero steady-state error (Integral action)
- Reduced overshoot (Derivative action)

Research shows that PID controllers significantly improve temperature stability in heating systems compared to ON/OFF control. Proper tuning of PID parameters ensures

smooth temperature rise and minimal fluctuation, making PID control ideal for milk heating and pasteurization processes.

2.8 Comparison of ON/OFF and PID Control

Several studies compare ON/OFF and PID control techniques and conclude that:

- ON/OFF control is simple but less accurate
- PID control provides smoother temperature regulation
- PID control reduces thermal stress on heating elements
- PID improves energy efficiency

Due to these advantages, modern dairy heating systems increasingly adopt PID-based control strategies.

2.9 Conclusion

Based on the literature reviewed:

- Accurate temperature control is essential in milk heating applications
- NTC thermistors are suitable for moderate temperature measurement
- STM32 microcontrollers provide reliable ADC and control capabilities
- ON/OFF control is simple but limited in performance
- PID control offers superior temperature stability and precision

These findings form the basis for the design and implementation of the proposed **Milk Temperature Monitoring and Control System using STM32**, as discussed in the subsequent chapters.

CHAPTER 3



CHAPTER 3

SYSTEM OVERVIEW AND BLOCK DIAGRAM DESCRIPTION

3.1 Introduction

This chapter describes the overall system architecture of the **Milk Temperature Monitoring and Control System**. It explains the working principle, block diagram, and interaction between hardware and software components. The system is designed to monitor milk temperature accurately and control the heating process using **ON/OFF control** and **PID control techniques** implemented on an STM32 microcontroller.

3.2 System Overview

The proposed system continuously measures the temperature of milk using an NTC thermistor sensor. The analog signal from the sensor is converted into digital form using the ADC module of the STM32 microcontroller. Based on the measured temperature and predefined setpoint, the controller operates the heater using either ON/OFF logic or PID control.

The system ensures:

- Accurate temperature measurement
- Safe and controlled heating of milk
- Reduced temperature fluctuations
- Efficient energy usage

3.3 Block Diagram Description

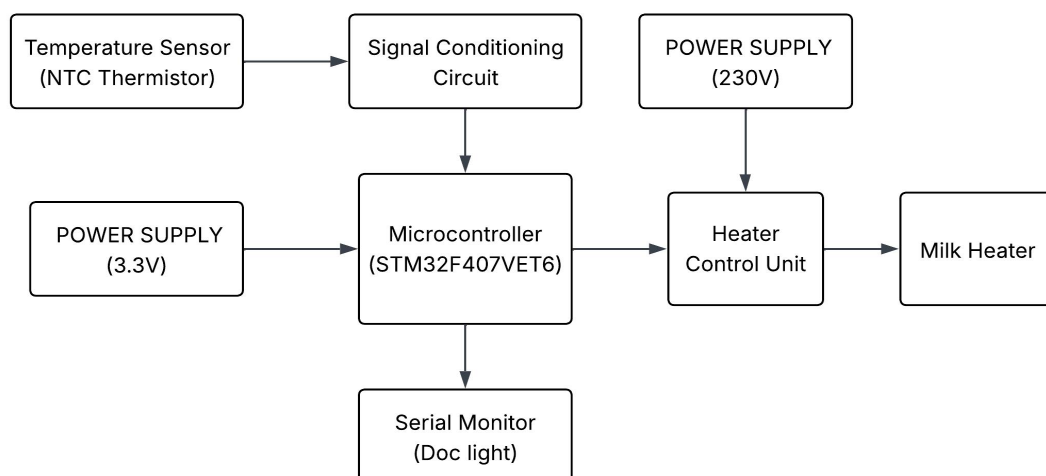


Fig.3.1 Block Diagram of Milk Heater

3.3.1 Power Supply Unit

The power supply unit provides regulated voltage required for the operation of the microcontroller and peripheral components. A stable **3.3 V supply** is used for the STM32 microcontroller and ADC reference. Proper decoupling capacitors are used to reduce noise and ensure reliable operation.

3.3.2 Temperature Sensor Unit

An **NTC thermistor (10 k Ω)** is used to sense the temperature of milk. As the temperature increases, the resistance of the thermistor decreases. The thermistor is connected in a voltage divider configuration to generate a voltage proportional to temperature.

The advantages of using an NTC thermistor include:

- High sensitivity in the operating temperature range
- Fast response time
- Low cost
- Ease of interfacing with ADC

3.3.3 Signal Conditioning Circuit

The voltage divider output from the thermistor is passed through a signal conditioning circuit consisting of:

- Series protection resistor
- RC low-pass filter
- Clamping diode for overvoltage protection

This circuit ensures:

- Noise reduction
- Protection of the ADC pin
- Stable and accurate ADC readings

3.3.4 STM32 Microcontroller Unit

The STM32 microcontroller acts as the core of the system. It performs the following functions:

- Reads analog voltage from the thermistor using ADC
- Converts ADC values into temperature using mathematical models

- Implements ON/OFF and PID control algorithms
- Controls the heater through GPIO and PWM outputs
- Transmits temperature data through UART for monitoring

The internal high-speed oscillator (HSI) is used as the clock source for system operation.

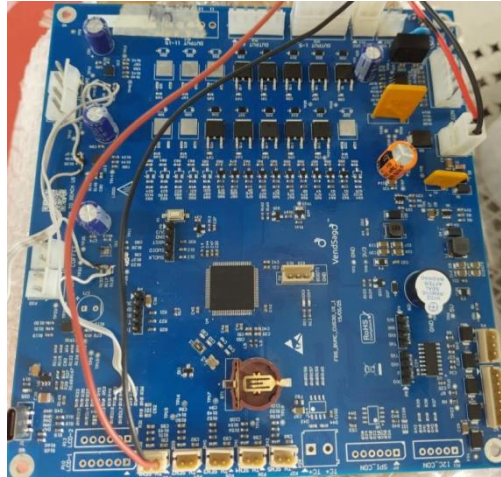


Fig.3.2 Microcontroller Unit

3.3.5 Heater Control Unit

The heater is controlled using a **MOSFET-based driver circuit**. The microcontroller output drives the MOSFET gate to switch the heater ON or OFF.

- In **ON/OFF control**, the heater is switched fully ON or OFF based on temperature comparison.
- In **PID control**, PWM signals are used to regulate the heater power smoothly.

This method improves temperature stability and reduces stress on the heating element.

3.4 Control Logic Overview

3.4.1 ON/OFF Control Logic

- Heater ON when temperature < setpoint
- Heater OFF when temperature $\geq$ setpoint
- Hysteresis is added to prevent frequent switching

This method is simple and suitable for basic milk heating applications.

3.4.2 PID Control Logic

PID control uses three components:

- **Proportional (P)** – Reacts to current error
- **Integral (I)** – Eliminates steady-state error
- **Derivative (D)** – Improves stability and response

The PID controller continuously adjusts heater power to maintain the milk temperature at the desired setpoint with minimal fluctuation.

3.5 System Operation Flow

1. Power ON the system
2. Initialize ADC, GPIO, UART, and Timer peripherals
3. Read thermistor voltage using ADC
4. Convert ADC value to temperature
5. Compare temperature with setpoint
6. Apply ON/OFF or PID control logic
7. Drive heater accordingly
8. Display or transmit temperature values
9. Repeat continuously

3.6 Advantages of the Proposed System

- Accurate milk temperature monitoring
- Improved temperature stability
- Reduced human intervention
- Flexible control using software
- Suitable for dairy and food-processing applications

3.7 Conclusion

This chapter presented the system overview and Block diagram description of the **Milk Temperature Monitoring and Control System using STM32**. The block diagram, sensor interfacing, control logic, and heater control methods were discussed in detail. The next chapter focuses on the **hardware module description** and individual components used in the system.

CHAPTER 4



CHAPTER 4

HARDWARE MODULE DESCRIPTION

4.1 Introduction

This chapter describes the hardware components used in the implementation of the **Milk Temperature Monitoring and Control System**. Each hardware module plays a vital role in ensuring accurate temperature sensing, reliable control, and safe operation of the heating system. The selection of components is based on accuracy, reliability, cost-effectiveness, and ease of integration with the STM32 microcontroller.

4.2 STM32 Microcontroller

The STM32 microcontroller is the core processing unit of the system. It is responsible for data acquisition, processing, control decision-making, and communication.

Features of STM32 Microcontroller

- 32-bit ARM Cortex-M core
- High-speed internal oscillator (HSI)
- 12-bit Analog-to-Digital Converter (ADC)
- Multiple GPIO pins
- Timer and PWM support
- UART communication interface
- Low power consumption

Role in the System

- Reads analog voltage from the thermistor using ADC
- Converts ADC values into temperature
- Executes ON/OFF and PID control algorithms
- Generates PWM signals for heater control
- Sends temperature data via UART for monitoring

4.3 Pinout of STM32F407VET6

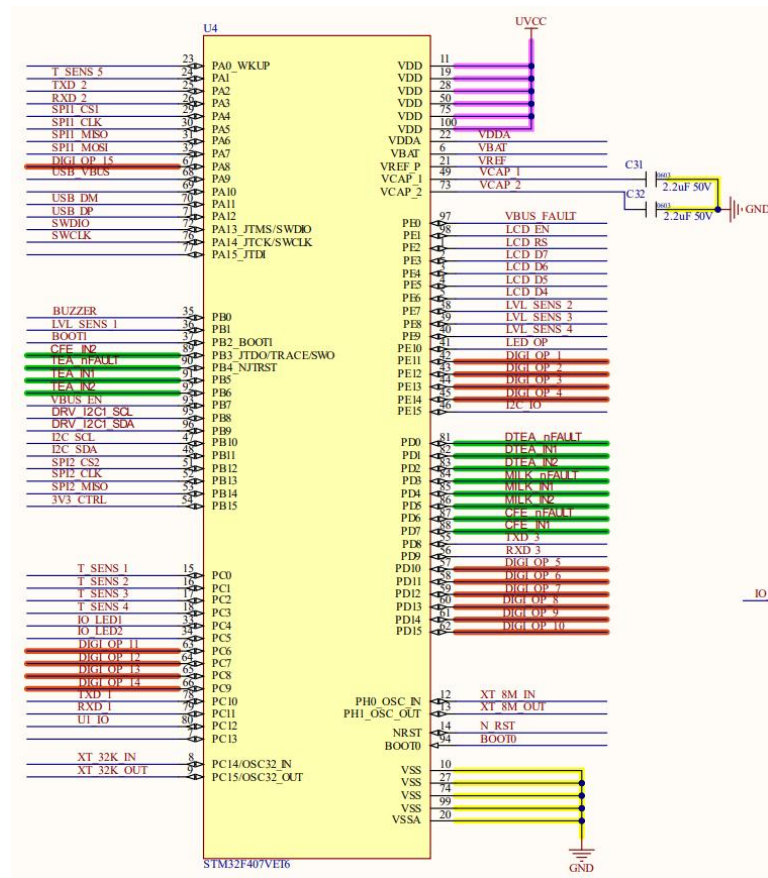


Fig.4.1 Pinout of STM32F407VET6

4.4 NTC Thermistor Sensor

An NTC (Negative Temperature Coefficient) thermistor with a nominal resistance of $10\text{ k}\Omega$ at 25°C is used to sense milk temperature.

Characteristics:

- Resistance decreases with increase in temperature
- High sensitivity in the temperature range of interest
- Fast thermal response
- Compact and cost-effective

Application in the System

The thermistor is immersed or thermally coupled with the milk container. Changes in milk temperature cause corresponding changes in thermistor resistance, which are used to determine temperature accurately.

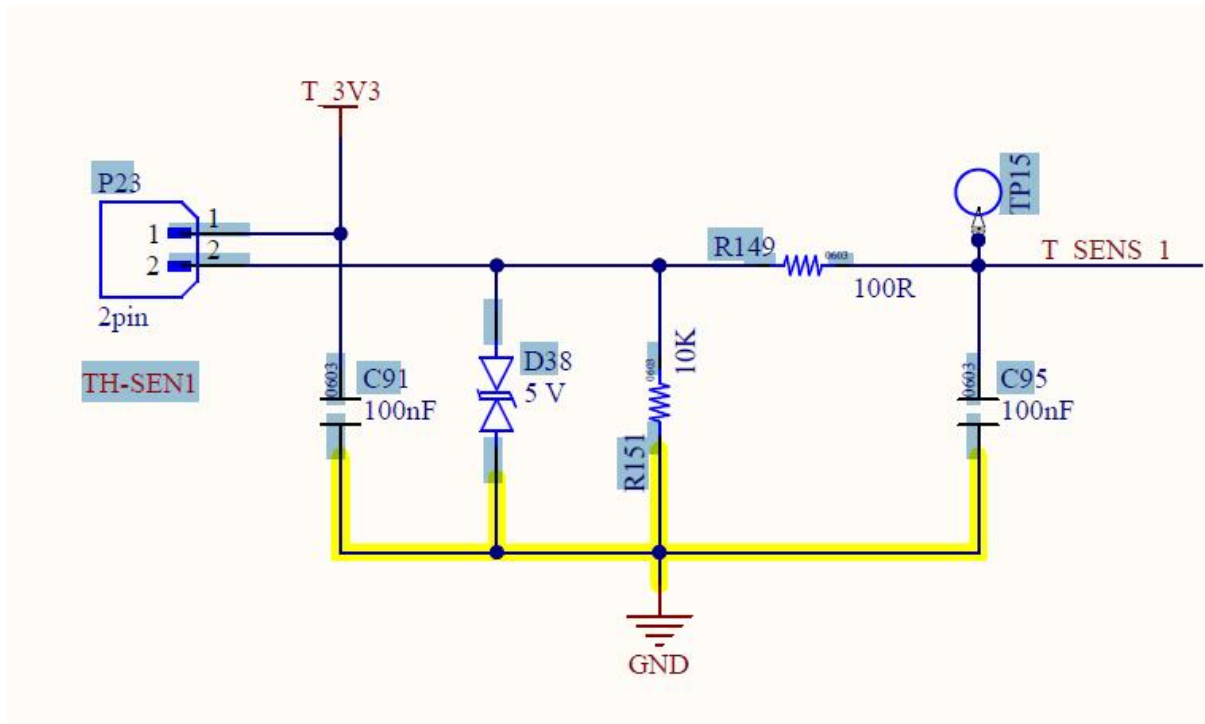


Fig.4.2 Circuit Diagram of Thermistor

4.5 Voltage Divider Network

The thermistor is connected in a voltage divider configuration along with a fixed resistor.

Purpose

- Converts resistance variation into voltage variation
- Produces an analog voltage suitable for ADC input

Working

As the thermistor resistance changes with temperature, the voltage across the divider changes proportionally. This voltage is fed into the ADC pin of the STM32 microcontroller.

4.6 Signal Conditioning Circuit

The signal conditioning circuit improves the quality and safety of the analog signal before it reaches the ADC.

Components Used

- Series resistor
- Low-pass RC filter

- Protection diode (if required)

Functions

- Reduces electrical noise
- Protects ADC from voltage spikes
- Improves measurement stability

4.7 Analog-to-Digital Converter (ADC)

The STM32's internal **12-bit ADC** converts the analog voltage from the thermistor circuit into a digital value ranging from 0 to 4095.

Key Features

- High resolution (12-bit)
- Programmable sampling time
- Software-triggered conversion

Importance

Accurate ADC conversion is essential for precise temperature measurement and control.

4.8 Heater Unit

The heater is the load used to raise the temperature of milk.

Specifications

- Electrical heating element
- Suitable power rating for milk heating
- Operates under controlled switching

Function

The heater increases the temperature of milk based on control commands from the microcontroller.

4.9 MOSFET-Based Driver Circuit

A power MOSFET is used to control the heater safely and efficiently.

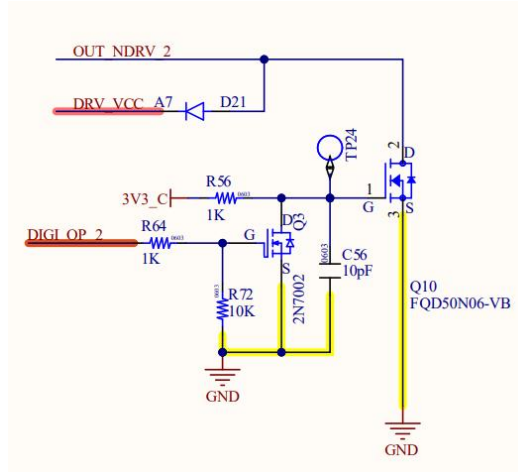


Fig.4.3 MOSFET Driver Circuit

Why MOSFET is Used?

- High current handling capability
- Fast switching
- Low power loss
- Electrical isolation between controller and heater

Operation

- Gate controlled by STM32 GPIO or PWM output
- Heater turns ON or OFF based on control logic
- Supports PWM-based power control in PID mode

4.10 Solid-State Switching

In some configurations, a relay or solid-state relay (SSR) may be used instead of or along with a MOSFET.

Purpose

- Electrical isolation
- Safe switching of high-power loads



Fig.4.4 Solid State Relay

4.11 UART Communication Module

UART communication is used for serial monitoring and debugging.

Functions

- Displays real-time temperature values
- Shows system status messages
- Helps in tuning PID parameters

Advantages

- Simple implementation
- Useful for testing and calibration
- Supports external USB-to-TTL converters

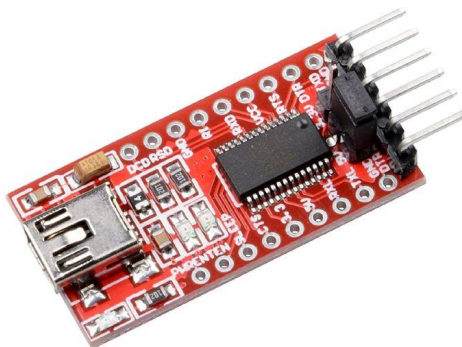


Fig.4.5 USB to TTL Converter

4.12 Indicator Components

Optional indicators such as LEDs can be used to indicate:

- Power ON status
- Heater ON/OFF status

- Fault conditions

4.13 Hardware Interfacing Summary

Component	Interfaced With
Thermistor	ADC pin of STM32
Heater	MOSFET driver circuit
MOSFET Gate	GPIO / PWM pin
UART Module	UART TX/RX pins
Power Supply	All system modules

Table.4.1 Hardware Interface

4.14 Safety Considerations

- Proper insulation for heater terminals
- Stable power supply for microcontroller
- Protection against overcurrent and overheating
- Secure grounding of all components

4.15 Conclusion

This chapter presented a detailed description of all hardware components used in the **Milk Temperature Monitoring and Control System**. Each module's function and importance were discussed. The next chapter explains the **software design and control algorithms** implemented in the STM32 microcontroller.

CHAPTER 5



CHAPTER 5

SOFTWARE DESIGN AND IMPLEMENTATION

5.1 Introduction

This chapter describes the software design and implementation of the **Milk Temperature Monitoring and Control System using STM32**. The software is developed using **STM32CubeMX** for peripheral configuration and **STM32CubeIDE** for program development. The embedded C program is responsible for temperature measurement, control logic execution, and real-time monitoring.

5.2 Development Tools Used

The following software tools were used in the development process:

- **STM32CubeMX** – Used for microcontroller selection and peripheral configuration
- **STM32CubeIDE** – Used for coding, compiling, and debugging
- **Embedded C** – Programming language
- **HAL (Hardware Abstraction Layer)** – For peripheral access
- **UART Serial Monitor** – For temperature display and debugging

5.3 Software Architecture

The software architecture follows a modular approach. Each module is designed to perform a specific function.

Major Software Modules

- System Clock Configuration
- ADC Initialization and Temperature Reading
- Thermistor Signal Processing
- Control Algorithm (ON/OFF and PID)
- PWM Generation
- UART Communication

5.4 Peripheral Configuration Using STM32CubeMX

5.4.1 ADC Configuration

- ADC Resolution: 12-bit
- Input Channel: Connected to NTC thermistor
- Conversion Mode: Single conversion
- Sampling Time: 144 cycles (to reduce noise)

The ADC converts the analog voltage from the thermistor circuit into digital values.

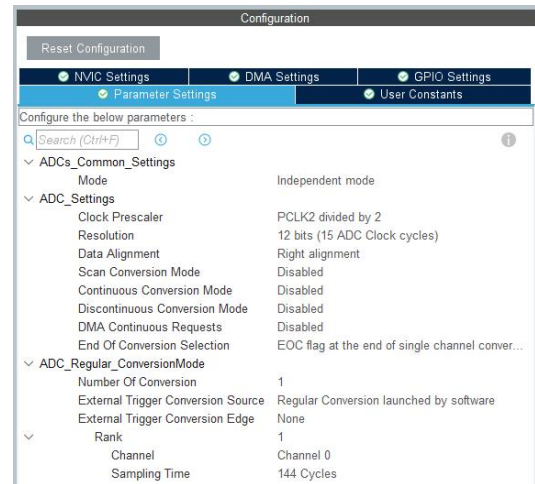


Fig.5.1 ADC Configuration in STM Cube MX

5.4.2 UART Configuration

- Baud Rate: 115200 bps
- Data Bits: 8
- Stop Bits: 1
- Parity: None
- Mode: Transmit and Receive

UART is used to display real-time temperature, error values, and control output.



Fig.5.2 UART Configuration in STM Cube MX

5.4.3 Timer and PWM Configuration

- Timer Used: TIM3 / TIM2
- Mode: PWM mode
- PWM Frequency: Suitable for heater control
- Duty Cycle: Controlled by PID output

PWM is used to regulate heater power smoothly.

5.4.4 GPIO Configuration

Pin	Function
PE12	Heater Control (Active LOW)
PB15	MOSFET Enable
PD10	Milk Valve Control
ADC Pin	Thermistor Input

Table.5.1 GPIO Configuration

5.5 Temperature Measurement Algorithm

The temperature measurement process involves the following steps:

1. Read ADC value from thermistor
2. Convert ADC value to voltage
3. Calculate thermistor resistance
4. Apply **Steinhart–Hart equation**
5. Convert Kelvin to Celsius
6. Display temperature via UART

Steinhart–Hart Equation

$$\frac{1}{T} = A + B \ln(R) + C(\ln(R))^3$$

This method provides higher accuracy over a wide temperature range.

5.6 Control Logic Implementation

5.6.1 ON/OFF Control Logic

If Temperature < Setpoint → Heater ON

If Temperature ≥ Setpoint → Heater OFF

This method is simple but causes oscillations near the setpoint.

5.6.2 PID Control Logic

PID control uses the following equation:

$$PID = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$$

Where:

- $e(t)$ is the temperature error
- K_p controls proportional response
- K_i eliminates steady-state error
- K_d improves stability

The PID output controls the PWM duty cycle.

5.7 Software Flowchart

Software Execution Flow:

1. System Initialization
2. Peripheral Initialization
3. ADC Reading
4. Temperature Calculation
5. Control Algorithm Execution
6. PWM Update
7. UART Data Transmission
8. Loop Continuation

5.8 Error Handling and Safety Features

- ADC range checking to detect sensor faults
- Heater OFF during abnormal readings
- Integral windup prevention in PID
- Default safe states during startup

5.9 Testing and Debugging

- Temperature values verified using external thermometer
- UART output monitored for debugging
- PID parameters tuned experimentally
- ON/OFF control behaviour validated

5.10 Conclusion

This chapter detailed the complete software design and implementation of the milk temperature monitoring and control system. The modular design ensures flexibility, reliability, and scalability. Both ON/OFF and PID control techniques were successfully implemented using STM32.

CHAPTER 6



CHAPTER 6

OUTPUT AND RESULT ANALYSIS

6.1 Introduction

This chapter presents the experimental results obtained from the implementation of the **Milk Temperature Monitoring and Control System using STM32**. The system performance was analysed under real operating conditions using both **ON/OFF control** and **PID control techniques**. The measured temperature values were compared with actual milk temperature to verify accuracy and stability.

6.2 Temperature Monitoring Results

The NTC thermistor was connected to milk container externally and connected to the STM32 ADC. The temperature was continuously monitored and displayed through UART communication.

The system successfully measured temperature variations in real time, and the values were found to closely match the actual milk temperature after proper calibration.

Observation:

- Temperature readings updated continuously
- Minor deviation observed due to sensor tolerance
- Accuracy improved after averaging and filtering

ADC: 1934	Temp: 22.27 C<CR><LF>	ADC: 1963	Temp: 22.97 C<CR><LF>
ADC: 1945	Temp: 22.54 C<CR><LF>	ADC: 1943	Temp: 22.49 C<CR><LF>
ADC: 1938	Temp: 22.37 C<CR><LF>	ADC: 1964	Temp: 22.99 C<CR><LF>
ADC: 1941	Temp: 22.44 C<CR><LF>	ADC: 1932	Temp: 22.23 C<CR><LF>
ADC: 1929	Temp: 22.15 C<CR><LF>	ADC: 1962	Temp: 22.95 C<CR><LF>
ADC: 1960	Temp: 22.90 C<CR><LF>	ADC: 1958	Temp: 22.85 C<CR><LF>
ADC: 1954	Temp: 22.75 C<CR><LF>	ADC: 1942	Temp: 22.47 C<CR><LF>
ADC: 1954	Temp: 22.75 C<CR><LF>	ADC: 1959	Temp: 22.87 C<CR><LF>
ADC: 1940	Temp: 22.42 C<CR><LF>	ADC: 1934	Temp: 22.27 C<CR><LF>
ADC: 1927	Temp: 22.11 C<CR><LF>	ADC: 1939	Temp: 22.39 C<CR><LF>
ADC: 1952	Temp: 22.71 C<CR><LF>	ADC: 1951	Temp: 22.68 C<CR><LF>
ADC: 1937	Temp: 22.35 C<CR><LF>	ADC: 1950	Temp: 22.66 C<CR><LF>
ADC: 1944	Temp: 22.51 C<CR><LF>	ADC: 1934	Temp: 22.27 C<CR><LF>
		ADC: 1939	Temp: 22.39 C<CR><LF>
		ADC: 1924	Temp: 22.03 C<CR><LF>

Fig.6.1 Temperature Output in Doc Light

6.3 ON/OFF Control Results

In ON/OFF control mode, the heater was switched **ON** when the milk temperature was below the setpoint and switched **OFF** when the setpoint was reached.

Results observed:

- Simple and reliable operation
- Temperature oscillations around setpoint
- Frequent heater switching at boundary conditions

Conclusion:

ON/OFF control is suitable for basic milk heating applications but lacks smooth temperature regulation.



Fig.6.2 ON/OFF System's Output

6.4 PID Control Results

PID control was implemented to improve system performance. The heater power was adjusted using PWM based on the PID output.

Results observed:

- Smooth rise in milk temperature
- Reduced overshoot
- Stable temperature near setpoint
- Improved heating efficiency

ASCII	HEX	Decimal	Binary	
Temp: 23.48 C	Error: 36.32 C	Corrected: 100.00<CR><LF>		
Temp: 24.10 C	Error: 35.90 C	Corrected: 100.00<CR><LF>		
Temp: 23.75 C	Error: 36.25 C	Corrected: 100.00<CR><LF>		
Temp: 23.82 C	Error: 36.18 C	Corrected: 100.00<CR><LF>		
Temp: 24.07 C	Error: 35.93 C	Corrected: 100.00<CR><LF>		
Temp: 24.37 C	Error: 35.63 C	Corrected: 100.00<CR><LF>		
Temp: 23.43 C	Error: 36.57 C	Corrected: 100.00<CR><LF>		
Temp: 23.40 C	Error: 36.60 C	Corrected: 100.00<CR><LF>		
Temp: 23.28 C	Error: 36.72 C	Corrected: 100.00<CR><LF>		
Temp: 22.99 C	Error: 37.01 C	Corrected: 100.00<CR><LF>		
Temp: 23.31 C	Error: 36.69 C	Corrected: 100.00<CR><LF>		
Temp: 24.00 C	Error: 36.00 C	Corrected: 100.00<CR><LF>		
Temp: 23.26 C	Error: 36.74 C	Corrected: 100.00<CR><LF>		
Temp: 23.53 C	Error: 36.47 C	Corrected: 100.00<CR><LF>		
Temp: 23.33 C	Error: 36.67 C	Corrected: 100.00<CR><LF>		
Temp: 23.45 C	Error: 36.55 C	Corrected: 100.00<CR><LF>		
Temp: 22.94 C	Error: 37.06 C	Corrected: 100.00<CR><LF>		
Temp: 24.07 C	Error: 35.93 C	Corrected: 100.00<CR><LF>		
Temp: 23.60 C	Error: 36.40 C	Corrected: 100.00<CR><LF>		
Temp: 23.45 C	Error: 36.55 C	Corrected: 100.00<CR><LF>		
Temp: 23.82 C	Error: 36.18 C	Corrected: 100.00<CR><LF>		
Temp: 23.13 C	Error: 36.87 C	Corrected: 100.00<CR><LF>		
Temp: 22.91 C	Error: 37.09 C	Corrected: 100.00<CR><LF>		
Temp: 23.40 C	Error: 36.60 C	Corrected: 100.00<CR><LF>		
Temp: 22.86 C	Error: 37.14 C	Corrected: 100.00<CR><LF>		
Temp: 23.63 C	Error: 36.37 C	Corrected: 100.00<CR><LF>		
Temp: 23.65 C	Error: 36.35 C	Corrected: 100.00<CR><LF>		
Temp: 23.80 C	Error: 36.20 C	Corrected: 100.00<CR><LF>		

Fig.6.3 PID Temperature Output in Doc Light

Conclusion:

PID control provided better stability and accuracy compared to ON/OFF control.

6.5 Comparison of ON/OFF and PID Control

Parameter	ON/OFF Control	PID Control
Temperature Stability	Low	High
Overshoot	High	Minimal
Heater Switching	Frequent	Smooth
Accuracy	Moderate	High

Table 6.1 Comparison of ON/OFF and PID Control

6.6 Overall System Performance

The system performed reliably under different temperature conditions. The STM32 microcontroller efficiently handled temperature sensing, control logic, and heater operation. The UART output enabled easy monitoring and debugging.

6.7 Conclusion

The experimental results confirm that the designed system is capable of accurate milk temperature monitoring and effective control. While ON/OFF control provides a simple solution, PID control significantly enhances system performance, making it suitable for practical dairy heating applications.

CHAPTER 7



CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

In this project, a **Milk Temperature Monitoring and Control System using STM32** was successfully designed and implemented. The system was capable of accurately measuring milk temperature using an NTC thermistor and controlling the heating process through both **ON/OFF control** and **PID control techniques**.

The ON/OFF control method demonstrated simple and reliable operation, suitable for basic heating applications. However, temperature fluctuations were observed near the setpoint. To overcome this limitation, a **PID control algorithm** was implemented, which provided smooth temperature regulation, reduced overshoot, and improved stability.

The STM32 microcontroller proved to be an efficient and reliable platform for real-time temperature control applications. The system achieved the desired performance and met the objectives of accurate temperature measurement, effective heater control, and safe operation.

7.2 Achievements of the Project

- Successful measurement of milk temperature using an NTC thermistor
- Implementation of ON/OFF temperature control logic
- Implementation of PID-based temperature control for improved stability
- Real-time monitoring using UART communication
- Reliable heater and valve control using GPIO and PWM

7.3 Limitations

- Temperature accuracy depends on thermistor tolerance and calibration
- ON/OFF control causes frequent switching near the setpoint
- Manual tuning of PID parameters is required
- System does not include automatic fault detection

7.4 Future Scope

The proposed system can be further enhanced with the following improvements:

1. **Automatic PID tuning** to optimize performance under varying conditions
2. **LCD or touchscreen display** for standalone operation
3. **Wireless monitoring** using IoT platforms (Wi-Fi / Bluetooth)
4. **Multiple temperature sensors** for large milk containers
5. **Data logging** for quality analysis in dairy processing
6. **Safety features** such as over-temperature protection and alarms

7.5 Final Remark

This project demonstrates a practical application of embedded systems in dairy processing. The developed system provides an efficient and reliable solution for milk temperature control and can be extended for industrial applications with further enhancements.